

NFL PASS-PROTECTION ANALYTICS

Pocket Integrity Score

The Complete Beginner's Guide

Everything in this app, explained in plain English:
football basics, every metric, every screen, and the data behind it.

Data: NFL Big Data Bowl 2023 (2021 season, weeks 1-8)

1. Start here: what this app is about

This app studies one specific part of American football: **pass protection**. When a team wants to throw the ball, the quarterback (QB) stands behind a wall of blockers. The defense tries to break through that wall and reach the QB before he can throw. The protected space around the QB is called the **pocket**.

The whole app is built to answer one question a real coach or scout would ask:

Which blockers build the cleanest, longest-lasting pockets, and which defenders break pockets down most consistently?

To answer it, we invented a score called the **Pocket Integrity Score (PIS)**. Think of it as a 0-to-100 grade for how safe the quarterback was on each pass play. 100 means a perfectly clean, roomy pocket; a low number means the QB was crowded, chased, or sacked.

You do not need to know any football to use this guide. We will build up every term from scratch.

2. Football basics (zero assumed knowledge)

The field

A football field is 120 yards long and about 53 yards wide. The middle 100 yards is the playing area; the two 10-yard strips at each end are the **end zones** (where you score). In this app we always flip plays so the offense is attacking **left to right**, so every play looks the same direction no matter which way it really went.

The two teams on a passing play

On every play there are two sides:

- **Offense** - the team with the ball, trying to move forward and score.
- **Defense** - the team trying to stop them.

A **pass play** is simply a play where the offense tries to throw the ball to a teammate rather than run with it.

Offensive positions (the people in gold in the app)

Short code	Position	What they do
QB	Quarterback	Throws the ball. The player we are protecting.
T	Tackle	Blocker on the far left/right end of the line. Protects the edges.
G	Guard	Blocker just inside the tackles.
C	Center	Blocker in the middle; snaps (hikes) the ball to the QB.
TE	Tight End	Hybrid: sometimes blocks, sometimes catches passes.
RB	Running Back	Usually runs/catches, but often stays in to block.
FB	Fullback	A heavier back who mostly blocks.

The five main blockers (two tackles, two guards, one center) are together called the **offensive line** or **O-line**. Left-to-right they line up as **LT, LG, C, RG, RT**.

Defensive positions (the people in red in the app)

Short code	Position	What they do
DE	Defensive End	Rushes the QB from the edge of the line.
DT	Defensive Tackle	Rushes the QB up the middle (interior).
NT	Nose Tackle	A DT lined up right over the center.
OLB	Outside Linebacker	Versatile; often rushes off the edge.
ILB / MLB	Inside / Middle Linebacker	Plays behind the line; rushes or covers.

Short code	Position	What they do
CB	Cornerback	Covers receivers, usually out wide.
S (FS/SS)	Safety (Free/Strong)	Covers deep, the last line of defense.

Defenders who try to reach the QB are called **pass rushers**. Defenders who guard receivers instead are in **coverage**.

The pocket, step by step

- The ball is **snapped** (hiked) to the QB - the play begins.
- The QB drops back a few steps; the blockers form a U-shaped wall in front and around him. That U-shape is the **pocket**.
- Pass rushers try to get around or through the blockers.
- The play ends when the QB **throws** (releases) the ball, is **sacked** (tackled before throwing), or **scrambles** (runs).

A clean pocket gives the QB time and space to throw accurately. A collapsing pocket forces rushed, inaccurate throws, sacks, or scrambles. PIS measures exactly this.

3. Key events on a play

The tracking data labels special moments. The app uses these to know when to start and stop measuring the pocket:

Event	Plain meaning
Snap (ball_snap)	The ball is hiked; the play starts. PIS starts measuring here.
Pass forward	The QB releases the ball. On a normal pass, PIS stops here.
QB sack	The QB was tackled before throwing. PIS stops here on a sack.
Scramble (run)	The QB gave up on throwing and ran. PIS stops here on a scramble.
Play action	A fake handoff to fool the defense before passing.
Pass arrived / caught / incomplete	What happened to the thrown ball.

The window from snap to the throw/sack/scramble is the only part of the play PIS cares about - that is when protection matters.

4. Pass-play outcomes (the 'result' filter)

Every pass play ends in one of five recorded results. You will see these codes in the Play Explorer and filters:

Code	Meaning	Good for the offense?
C	Complete pass (caught by a teammate)	Yes
I	Incomplete pass (hit the ground)	No
S	Sack (QB tackled before throwing)	Very bad
R	Scramble (QB ran instead of throwing)	Mixed
IN	Interception (caught by the defense)	Worst

Important idea: a completion can still have TERRIBLE protection (the QB got lucky), and a sack can happen behind GOOD protection (nobody was open, so the QB held the ball too long). That is why we measure the pocket itself, not just the result.

5. Formations (how the offense lines up)

A **formation** is the shape the offense makes before the snap. It hints at whether they will run or pass and how much protection the QB has. These are the ones in the app:

Formation	What it looks like
SHOTGUN	QB stands ~5 yards back to see the field better. The most common passing formation.
EMPTY	No running backs behind the QB - everyone is out as a receiver. Fewer blockers, riskier protection.
SINGLEBACK	One running back behind the QB. Balanced run/pass look.
I_FORM	Two backs lined up in a row behind the QB (an 'I' shape). A traditional, run-friendly look.
PISTOL	A shorter shotgun with a back directly behind the QB.
JUMBO	Extra big blockers in, almost no receivers. Maximum protection, usually short-yardage.
WILDCAT	A trick look where a non-QB takes the snap. Rare.

6. Dropback types (how the QB sets up to throw)

Dropback type	Plain meaning
TRADITIONAL	QB drops straight back and throws from the pocket. The standard.
SCRAMBLE	The pocket broke down, so the QB took off running.
DESIGNED_ROLLOUT_LEFT/RIGHT	The play was DESIGNED for the QB to move sideways to one side before throwing.
SCRAMBLE_ROLLOUT_LEFT/RIGHT	The QB improvised a run to the side under pressure.
DESIGNED_RUN	A planned QB run (not really a pass).
UNKNOWN	The data could not classify it.

Rollouts and scrambles move the QB out of the normal pocket, so PIS handles them a little differently (it credits a QB who escapes trouble rather than blaming the pocket).

7. Pass coverages (how the defense guards receivers)

Coverage is the defense's plan for guarding the receivers while (or instead of) rushing the QB. You do not need these to understand PIS, but they appear in filters, so here is the plain version. The number usually tells you how many defenders are guarding the deep part of the field.

Coverage	Simple idea
Cover-0	Everyone guards a man, NOBODY deep. All-out aggression, often a blitz. Hardest on protection.
Cover-1	One deep safety; everyone else guards a man.
Cover-2	Two deep safeties split the deep field.
Cover-3	Three defenders split the deep field. Very common.
Cover-6 / Quarters	The deep field split into quarters by 2-4 defenders.
2-Man	Two deep safeties, man coverage underneath.
Bracket	Two defenders double-team one dangerous receiver.
Prevent	Soft, deep coverage to stop a long play at the end of a half.
Red Zone / Goal Line	Special packages near the end zone.

Man coverage means a defender follows one specific receiver. **Zone coverage** means a defender guards an area of the field. **Blitz** means the defense sends extra rushers at the QB - great for pressure, risky because fewer defenders are left in coverage.

8. The star of the show: Pocket Integrity Score (PIS)

PIS is a 0-100 grade of how well the offense protected the QB on a single pass play. Higher is better protection.

How to read a PIS number

PIS range	What it means	Colour in app
85 - 100	Pristine pocket; QB had all day	Green
70 - 85	Solid protection	Light green
55 - 70	Shaky; pressure was building	Yellow
40 - 55	Poor; QB under real duress	Orange
0 - 40	Pocket collapsed; sack or heavy pressure	Red

What goes into the score

PIS blends five things we measure every tenth of a second from the player tracking data, each phrased as 'bigger is better':

Ingredient (weight)	Plain-English question it answers
Clean-space sustain (30%)	On average, how much open space did the QB have from the nearest rusher?
Pressure load (20%)	How few rushers got right up in the QB's face?
Collapse resistance (20%)	Did the pocket keep its size and depth, or shrink toward the QB?
Closing control (15%)	How well did blockers slow down rushers charging at the QB?
Last-second integrity (15%)	Was the pocket still clean in the final half-second before the throw?

Those five combine into a 'geometry score'. Then we nudge it a little by what actually happened, so the number agrees with reality:

Outcome	Adjustment to PIS
Sack	-12 points
QB was hit	-6 points
Hurry (QB rushed)	-4 points
Scramble	-3 points
Clean throw, no pressure	+2 points

The adjustment is small on purpose. PIS is driven mainly by the tracking geometry, not by the box score. That is what makes it more honest than just counting sacks.

If a play is missing too much data to measure fairly, we mark it 'unscorable' and leave it out rather than guessing. About 100% of plays in this dataset were scorable.

9. The two player ratings: PPPR and PCR

PPPR - Player Pocket Preservation Rating (for blockers)

A 0-100 rating of how well an **offensive blocker** (tackle, guard, center, tight end, back) protects the pocket. We take the pocket quality (PIS) on his plays and subtract penalties for the hits, hurries and sacks he personally allowed. Higher = better protector. It is a per-snap average, so a steady, reliable blocker beats one who has a few great plays and several disasters.

PCR - Pressure Creation Rating (for pass rushers)

A 0-100 rating of how consistently a **defender** disrupts pockets: how close and fast he gets to the QB plus the hits, hurries and sacks he produces. Higher = better rusher. Again it rewards doing it **consistently**, not one highlight sack.

Both ratings use a minimum snap count (default 50) so we only rank players with enough evidence, and each carries an uncertainty band (the '+/-' column) - wider means less certain.

10. The supporting stats (every column you'll see)

The pressure vocabulary

Term	Plain meaning
Sack	QB tackled behind the line before throwing. The worst outcome for protection.
Hit	A rusher hit the QB as or just after he threw.
Hurry	A rusher got close enough to rush/disrupt the throw.
Pressure	Any of the above (hit OR hurry OR sack). The catch-all.
Beaten	PFF judged that a blocker was clearly beaten by his defender.

For blockers these are 'allowed' (bad for them); for rushers they are 'created' (good for them). It is the same event seen from the two sides.

Leaderboard & table columns

Column	What it tells you
Snaps	How many plays the player was involved in. More = more reliable rating.
Avg PIS	Average pocket quality on that player's plays.
Press / Press%	Pressures (count) and the share of snaps with a pressure.
Hits / Hur / Sacks	Counts of each pressure type (allowed or created).
Def win% / Rush win%	In matchups: share of reps the defender/rusher won (got a pressure).
ProxImpact	A 0-1 tracking measure of how close and fast a rusher got to the QB.
MinDist	On average, the closest a rusher got to the QB each play, in yards.
Reps	In matchups: how many times a specific blocker and rusher faced each other.
+/- (bootstrap SD)	Uncertainty of the rating. Smaller = more confident.

11. The app, screen by screen

Home

The front page: top-line counts, a plain explanation of PIS, a validation summary (proof the metric is meaningful), and three ready-made example plays you can open with one click.

Play Explorer

The heart of the app. Pick a play and watch it animate on a football field.

- **Gold dots** = pass blockers. **Red dots** = pass rushers. **White dot** = the QB. **Grey dots** = receivers/coverage. **Brown diamond** = the ball.
- **Press Play** or drag the timeline slider to move through the play. The slider marks the snap and the end of the play.
- The shaded shape around the QB is the **pocket**. Watch it shrink when protection breaks down.
- The chart below shows PIS and the QB's open space second-by-second - you can literally see a collapse happen.

The layer toggles (left sidebar)

Layer	What it draws
Pocket shape	The U-shaped protected area around the QB.
QB clean-space ring	A circle showing the 'comfortable' space the QB wants.
Pressure radius	A small circle; rushers inside it are crowding the QB.
Player trails	Short tails showing where each player just came from.
Velocity arrows	Arrows showing each player's speed and direction.

Player Leaderboards

Two ranked tables: blockers by PPPR and rushers by PCR. Use the filters (week, team, position, formation, coverage, dropback, play action, minimum snaps) to narrow things down. Click a player button to jump to one of their plays in the Play Explorer.

Matchup Inspector

Head-to-head battles. Pick an offensive lineman to see which rushers he faced and how often he held up, or pick a rusher to see which blockers he beat. 'Win%' is how often the rusher got a pressure in that matchup. We only trust matchups with a few reps behind them.

Team / Unit View

The big picture. Rank every team's offensive line (who protects best) and every defense's front (who pressures best), and see how pocket quality changes by formation, coverage, down, and play action, with a plain-English takeaway on each chart.

12. Where the numbers come from (the data)

Everything is built from the NFL's Big Data Bowl 2023 dataset: the 2021 season, weeks 1-8, 122 games, 8,557 pass plays. Five files feed the app:

File	What's in it
games	One row per game: teams, week, date.
plays	One row per play: formation, coverage, result, down, etc.
players	One row per player: name, position, height, weight.
pffScoutingData	Expert PFF grades: each player's role and the pressures they created/allowed, plus who blocked whom.
tracking	The GPS-style data: every player's position 10 times per second. This is what powers the animation and PIS.

In the tracking data, the ball is a special row with no player ID. Players are matched across files by their IDs, and we carefully handle missing data, duplicate rows, and players running on or off the screen.

13. Honest limitations (what PIS can't see)

- PIS is a **top-down 2-D view**. It cannot see hand-fighting, leverage, or exactly who touched whom - it complements expert film and PFF grades, it does not replace them.
- **Eight weeks is a small sample**. Ratings for players with few snaps are uncertain (that is what the '+/-' band is for).
- **Scrambles are the hardest plays to measure** cleanly, so those are flagged.
- When the data does not say which blocker was assigned to which rusher, we mark the matchup 'uncertain' instead of guessing.

The golden rule: PIS tells you how clean the pocket was, which is strongly linked to pressure - but a link is not proof of cause. Use it as a smart starting point, then watch the play.

14. One-page cheat sheet

If you see...	Think...
PIS 90	Great protection, QB was comfortable.
PIS 40	Pocket collapsed, QB in trouble.
High PPR	A reliable, high-quality blocker.
High PCR	A consistent pressure-creating rusher.
Gold dot	A blocker (offense).
Red dot	A pass rusher (defense).
White dot	The quarterback.
'Sack' (S)	QB tackled before throwing - worst protection outcome.
'Pressure'	A hit, hurry, or sack - any disruption of the QB.
Cover-0 / blitz	Aggressive defense; expect pressure.
Shotgun / Empty	Passing looks; Empty has fewer blockers.
Wide '+/-' band	Rating is uncertain (few snaps).

That's everything. Keep this guide open next to the app, and within a few plays the terms will feel natural. Enjoy exploring the pocket!